

AHU screening (single-system, 2026-06)

Offline report from mapped roles and history. The same PDF is produced from a device folder, an Open-FDD server, or a future vendor adapter. This is not the BUILDING_100 Overview-mirrored lab PDF.

AHU_1

Sensor checks

Checks for stuck, out-of-range, or stale sensors. Only failed checks are listed when something is wrong.

Fail. Outdoor air temperature: reading out of physical range (low 35.0, high 200.0).

Fail. Return air temperature: sensor stuck flat; Supply air temperature: sensor stuck flat.

Fail. Outdoor air temperature: reading jumped suddenly.

Fail. Outdoor air temperature: reading changed faster than expected.

Anomaly screening

A quick look at whether each varying trend is ordinary this month. Not an equipment fault list.

Mixed air temperature: looks normal.

Outdoor air temperature: looks normal.

Executive summary

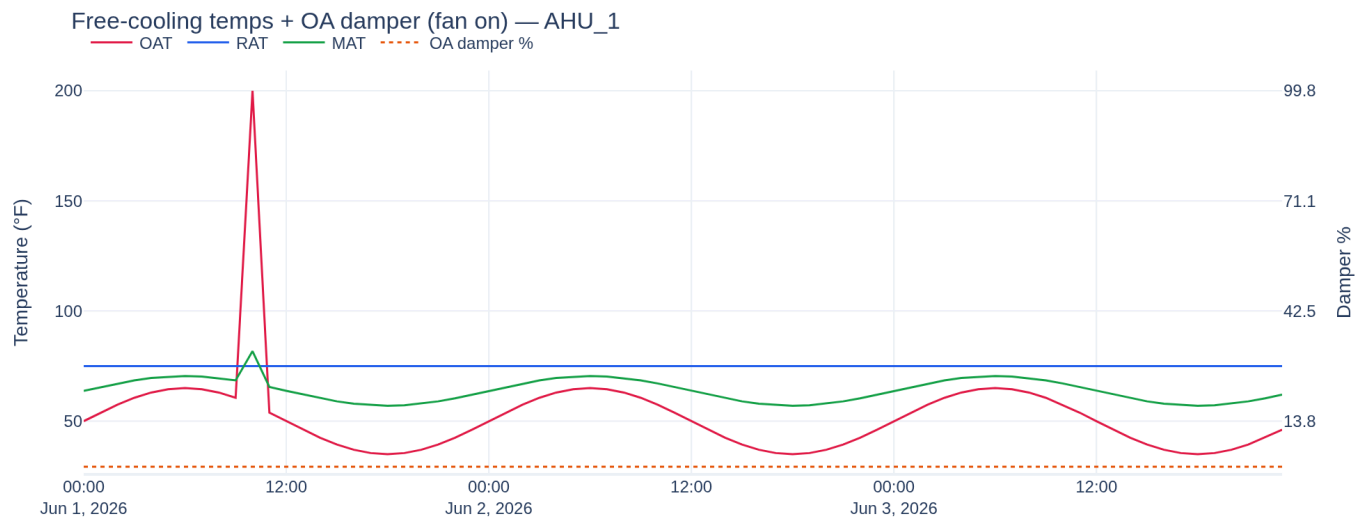
AHU_1 in June 2026 (72 samples, 72.0 h). Sensor checks found issues. Fail. Outdoor air temperature: reading out of physical range (low 35.0, high 200.0). Fail. Return air temperature: sensor stuck flat; Supply air temperature: sensor stuck flat. Fail. Outdoor air temperature: reading jumped suddenly. Fail. Outdoor air temperature: reading changed faster than expected. Anomaly screening: screened trends look normal. These screening notes are not equipment faults. Confirmed operating findings: Duct static below SP at full fan (GL36 A) 72.0 h; Economizer stuck closed 26.0 h; Always-on fan or pump runtime 72.0 h. The fan was on for 100.0 percent of the samples.

RCx week (2026-06-01 to 2026-06-03 UTC)

One week inside the filtered month. Figures follow the mapped roles for this system profile.

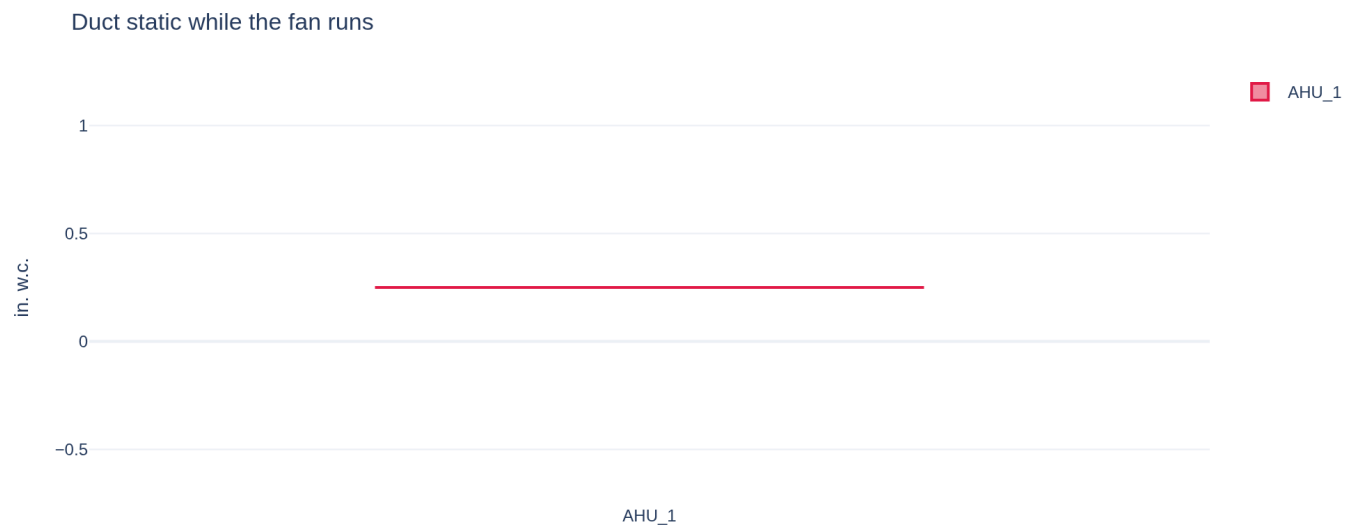
Free-cooling temperatures and outdoor-air damper

Outdoor, return, mixed, and supply air on the temperature axis. Damper percent is on the right axis. Fan running only.



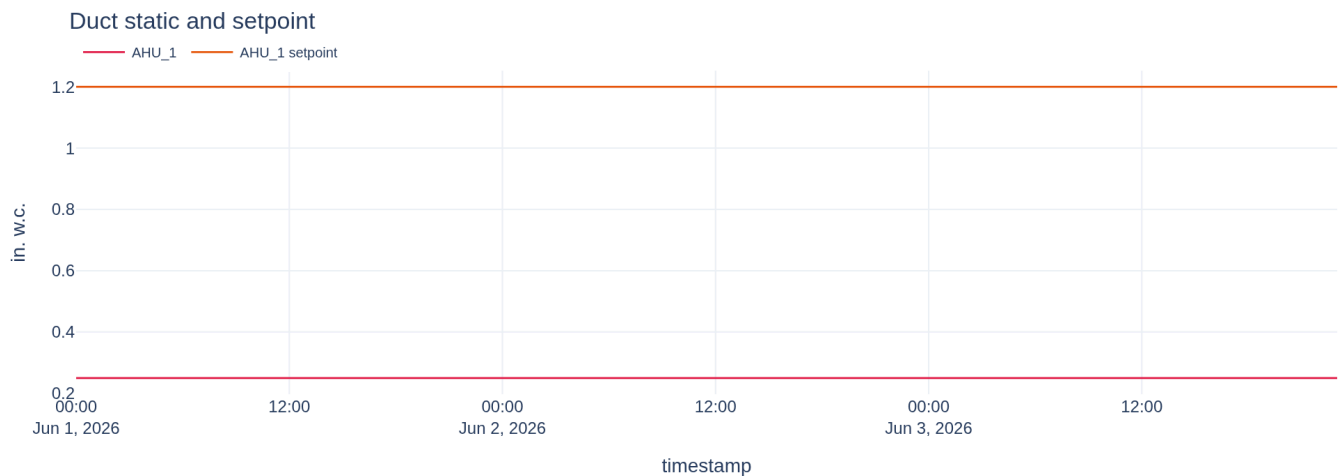
Duct static while the fan runs

Duct static pressure with the fan running.



Duct static and setpoint

Duct static pressure with the mapped setpoint.



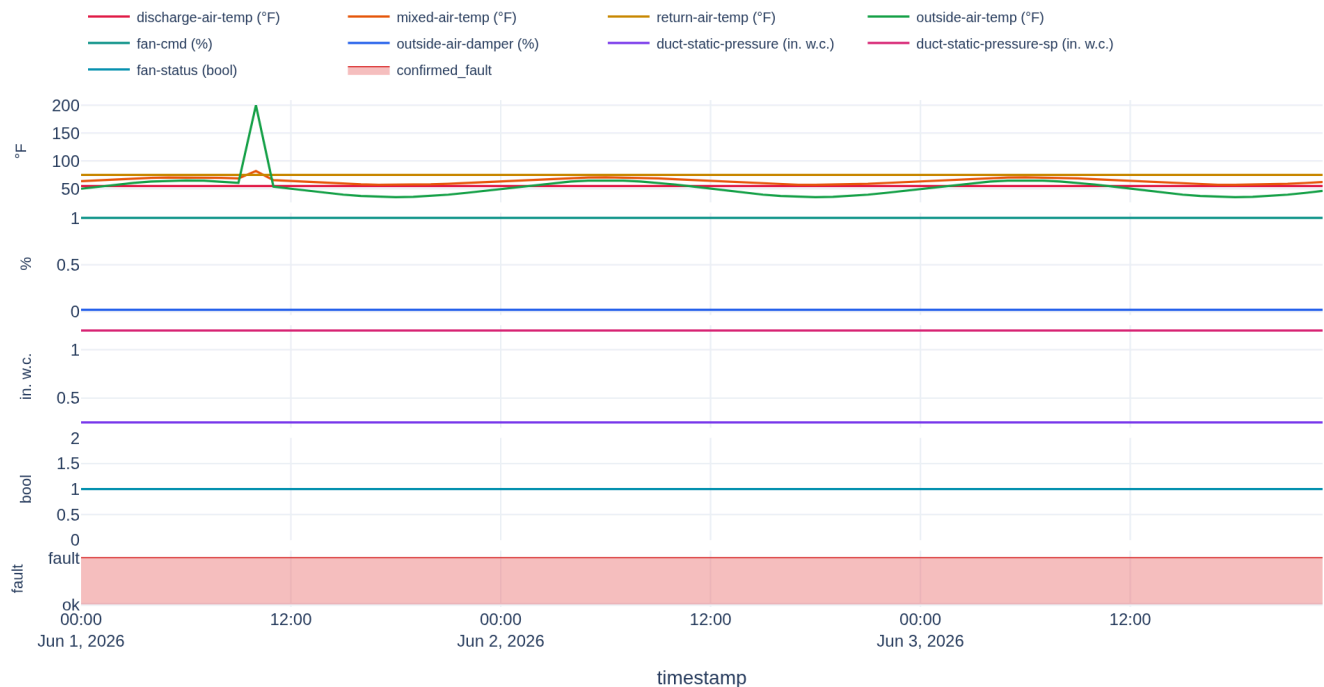
Confirmed faults

A figure is included only when the cookbook rule has confirmed fault hours in the filtered month.

FC1 – Duct static below SP at full fan (GL36 A)

Troubleshoot: Flags duct static too low while the supply fan is near full speed. Check: $DSP < DSPSP - \epsilon DSP$ AND $VFDSPD \geq 100\% - \epsilon VFDSPD$.

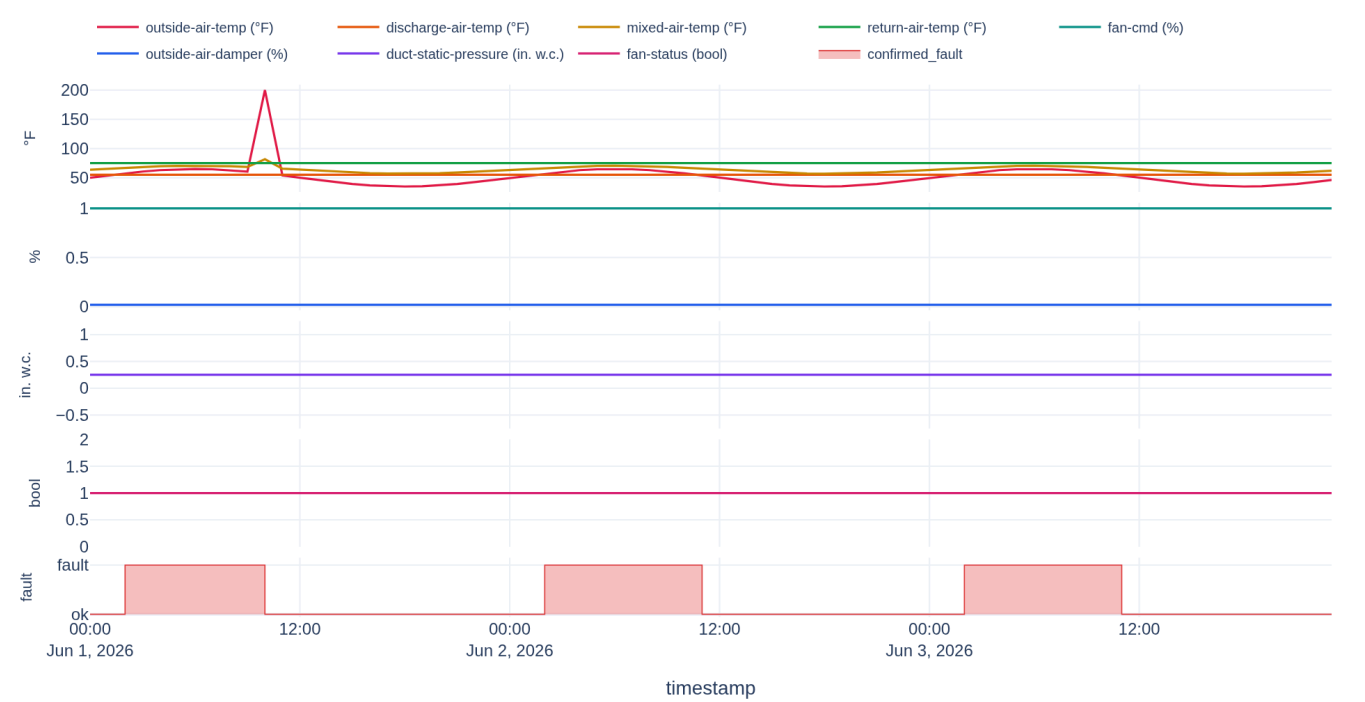
In the data: In June 2026 this holds for 72.0 hours, 2026-06-01 00:00 UTC through 2026-06-03 23:00 UTC. duct static pressure was on for 0.0 percent of those samples; duct static setpoint was on for 100.0 percent of those samples; fan speed command was on for 100.0 percent of those samples.



ECON-1 – Economizer stuck closed

Troubleshoot: Fan on, OA damper < 5%, OAT > 55°F (should be economizing). Check: Fan on, OA damper < 5%, OAT > 55°F (should be economizing).

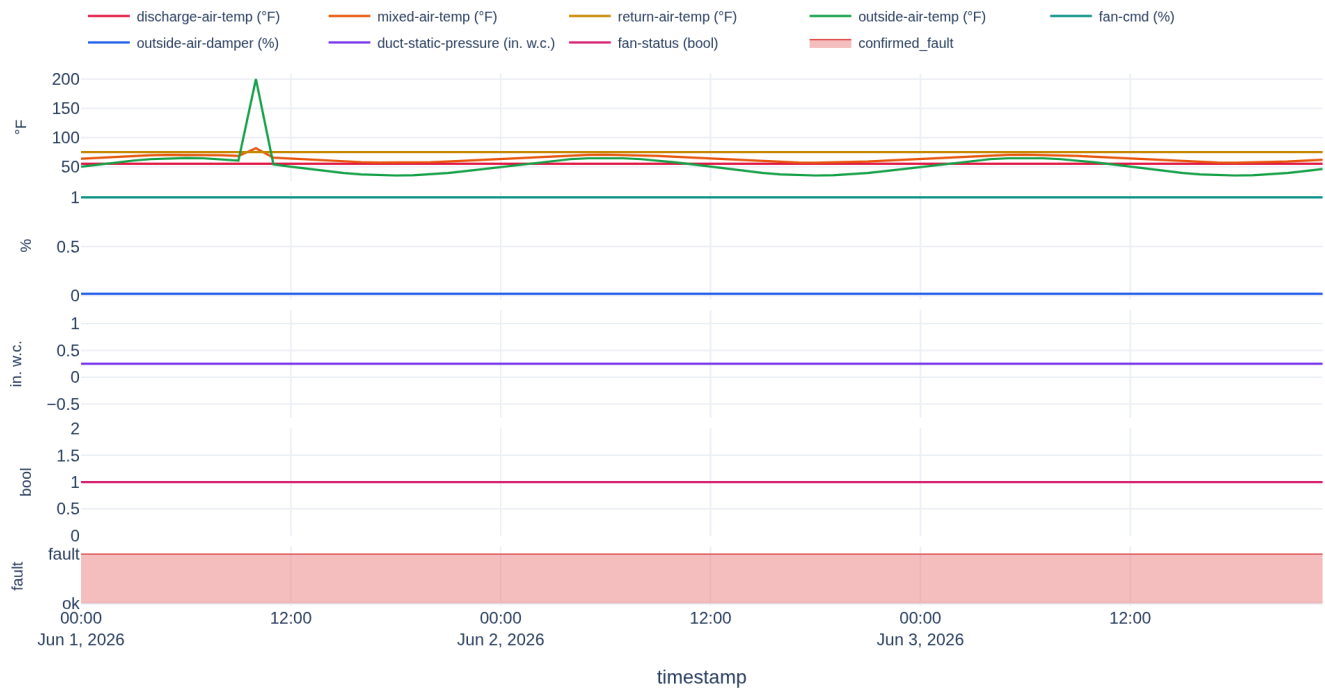
In the data: In June 2026 this holds for 26.0 hours, 2026-06-01 02:00 UTC through 2026-06-03 10:00 UTC. fan speed command was on for 100.0 percent of those samples; outdoor air damper was on for 0.0 percent of those samples; outdoor air temperature was about 63.0.



SCHED-247 – Always-on fan or pump runtime

Troubleshoot: Flags fans or pumps that stay on for nearly the entire analysis window (24/7 runtime). Check: Ranked proof (status/current > command) is on for ≥ always_on_pct of the window. Pressure is inferred runtime only and does not trip this rule ID.

In the data: In June 2026 this holds for 72.0 hours, 2026-06-01 00:00 UTC through 2026-06-03 23:00 UTC. fan status was on for 100.0 percent of those samples; fan speed command was on for 100.0 percent of those samples; duct static pressure was on for 0.0 percent of those samples.



Economizer delta scatter

economizer_delta_scatter: $x = \text{OAT} - \text{RAT} (\text{delta_or_f})$, $y = \text{MAT} - \text{RAT} (\text{delta_mr_f})$. Reference lines $y = \text{OA fraction} * x$ for 0/25/50/75/100% OA. Fan ON and $|\text{OAT} - \text{RAT}| \geq 10$ F. Viewport is the bottom-left mixing quadrant only (both deltas ≤ 0 : $\text{OAT} \leq \text{RAT}$ and $\text{MAT} \leq \text{RAT}$). Do not plot $\text{OAT} - \text{MAT}$ vs $\text{RAT} - \text{MAT}$.

